

Aaditya Kalwani

Aerospace Engineering student focused on machine learning applied to physical engineering systems. Currently researching battery state-of-charge estimation toward publication. Comfortable across the stack: neural networks from scratch in NumPy, C# object-oriented simulations and computational tools for constrained engineering problems. Hands-on UAV 'design and make' experience (Team Bath Drones).

Experience & Technical Projects

Undergraduate Researcher -- Battery State-of-Charge Estimation, Uni of Bath | Jun '26 - Present

- **Developing a first-principles machine learning model** to estimate lithium-ion battery State of Charge by fusing ultrasound acoustic sensing with coulomb counting, targeting journal publication
- **Implemented a multilayer perceptron from scratch** in NumPy (forward pass, hand-derived backpropagation, training loop) before adopting frameworks, to ground the model in first principles
- Addressed hysteresis, temperature, and current dependencies to generalise from lab charge-discharge cycles to real mission profiles using a high-fidelity ultrasound-SOC dataset

Aerodynamics Division Lead -- Team Bath Drones, Design & Make Competition | Nov '25 – Present

- **Engineered an autonomous UAV** under strict objectives and material constraints, and **successfully delivered the Critical Design Review**
- Optimised aerodynamic efficiency: evaluated Clark-Y and NACA0015 profiles for a 30% increase in lift-to-drag ratio; co-led the 'tadpole' fuselage configuration and CAD model

Fifth-9 (Intern) -- Tech Strategy & Product Development (Canary Wharf - Onsite) | Jul & Aug '25

- **Full Stack Development:** Architected and deployed 'PlayDay', a React/Node.js travel-planning engine, integrating the Google Gemini API for real-time, generative itineraries within a 5-day sprint
- **Strategic Analysis:** Formulated technical pitch decks and presented go-to-market strategies on emerging technologies (AI in Banking, Post-Quantum Cryptography) directly to company directors.

Private Tutor -- GCSE & A-Level STEM (ETutorYou and independent) | Sep '23 - Present

- Managed a private tutoring portfolio for 10+ STEM students: Improved student performance by >3 grades via **AI-automation** (Cowork), personalised feedback and targeted problem-set derivation.

Selected Engineering Projects

- Developed a 5:1 planetary transmission using **CAD**, complete with motion simulations in **Inventor**
- **Computational C# Tools:** Built an $n \times n$ matrix inverter utilising Gaussian elimination and cofactor matrices to mirror CFD pre-processing logic, alongside a console-based OOP simulator.

Education

BEng Aerospace Engineering (Pred. First-Class Honours), University of Bath, | Sep '25 - Present

- Relevant coursework: Thermofluids, Solid Mechanics, Mechatronics, Mathematics

Peter Symonds College, Winchester | Sep '23 - Jun '25

- **A-Level Results:** A*AAB | Mathematics, Physics, Further Maths, Computer Science
- 2x MEI National Maths Comp. | Silver, Senior Maths Challenge | Bronze, British Physics Olympiad |

Gordon's School, Woking | Sep '21 - Jun '23

- **GCSE Results:** 6x Grade 9s, including English & Further Maths | 5x Grade 8s
- **Founder & Leader of Coding Club** | Robotics and IOT Club | Silver - Maths & Physics Olympiads

Selected Technical Skills

- **Languages:** English, Hindi | Python (NumPy, Pandas, ML), C# (.NET/OOP), React, Node.js
- **Engineering Tools:** CAD (Autodesk Inventor), MATLAB, Git+GitHub